Group theory

Chapter 1. Groups

Exercise 1.1. Consider a

See Exercise 1.1.

Exercise 1.2. funn

Solve Exercise 1.2.

§1.1 Fields

We begin by establishing the properties of fields mapping vector components.

Definition 1.3. A function $T : V \to W$ between two vector spaces over the same field F is called a **linear transformation** if it satisfies:

1. Additivity: $T(u + v) = T(u) + T(v)$ for all $u, v \in V$
2. Homogeneity: $T(cv) = cT(v)$ for all $c \in F$ and $v \in V$

Definition 1.3.

Definition 1.4. …

§1.2 Degree of an extension

Theorem 1.5. Let V and W be vector spaces, where V is finite-dimensional. If $T : V \to W$ is a linear map, then:

$$\dim(\text{null } T) + \dim(\text{range } T) = \dim V$$

§1.3 Exercises

Exercise 1.6. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find a structured matrix representation M_T and verify the output dimension explicitly:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Exercise 1.7 (Identity Dimension Mapping).

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find the dimension of the null space.

Solution:

By inspecting the matrix row parameters:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The rank is clearly 2 because the two rows are linearly independent. By Rank-Nullity, $\dim(\text{null } T) = 3 - 2 = 1$.

Note 1.8. Remember that every basis is linearly independent.

Warning 1.9. Do **not** confuse image and codomain.

Example 1.10. Let

$$T(x, y) = (x + y, y).$$

Proof. We proceed by induction... □

Remark 1.11. The converse is false.

History 1.12. The Rank-Nullity theorem appeared in the nineteenth century.

Lemma 1.13. lemma statement

Proposition 1.14. proposition statement

Claim 1.15. claim statement

Corollary 1.16. corollary statement

Chapter 2. Symmetric groups

§2.1 Core Definitions

In this session, we analyze how linear transformations scale specific vector subspaces.

§2.1.1 some defn

Definition 2.1. Let $T : V \to V$ be a linear operator on a vector space V over a field F. A non-zero vector $v \in V$ is called an **eigenvector** of T if there exists a scalar $\lambda \in F$ such that:

$$T(v) = \lambda v$$

The scalar λ is called the **eigenvalue** corresponding to the eigenvector v.

See Definition 2.1 for the formal definition of eigenvalues and eigenvectors.

§2.2 Fundamental Theorems

Theorem 2.2. Let $T : V \to V$ be a linear operator. If $v_1, v_2, ..., v_m$ are eigenvectors corresponding to **distinct** eigenvalues $\lambda_1, \lambda_2, ..., \lambda_m$, then the set $\{v_1, v_2, ..., v_m\}$ is linearly independent.

§2.3 Homework Practice

Exercise 2.3. Consider the 2×2 matrix A representing a linear transformation on $\mathbb{R}^2$:

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$$

1. Compute the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$.
2. Find the distinct eigenvalues λ_1 and λ_2 by solving:

$$\lambda^2 - 7\lambda + 10 = 0$$

Chapter 3. Sylow's theorems

Chapter 4. Core Definitions

Theorem 3.1. There are infinitely many prime numbers.